Predicting Coronary Heart Disease from Survey Answers with Regularized Logistic Regression

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Abstract—

I. Introduction

The training set has 328 135 respondents and the test set 109 379, each described by 321 answers from a telephone health survey. The label is 1 if the respondent was diagnosed with MICHD (heart attack or angina), and 8.8% of the training labels are positive. Predicting the majority class gives 91% accuracy but an F1 of 0, so all models are compared by the F1 of the positive class, which is also the leaderboard metric.

II. Data and Feature Engineering

Most columns are not measurements but answer codes. For example, general health (GENHLTH) takes values 1 (excellent) to 5 (poor), plus 7 = "don't know" and 9 = "refused". A single weight on the raw value treats "refused" as worse than "poor". Many questions are also asked only to a subset of respondents: 147 of the 321 columns are missing in more than half of the rows.

We build the following features. All statistics (levels, quantiles, means) are computed on the training rows only, inside each cross-validation fold.

- Dropped: identifiers, interview dates, state, survey weights, constant columns and columns with less than 1% observed values.
- Categorical columns ($\leq$ 60 distinct values): one-hot encoding, with answer codes kept as their own levels. Levels seen in less than 0.2% of rows are merged into the reference level, which is dropped to avoid collinearity with the bias.
- Continuous columns: frequent answer codes (e.g. 7777, 9999) become indicators. The remaining values are clipped to their 1st and 99th percentiles, standardized, and expanded with a squared term (polynomial basis of degree 2).
- Missing indicators for every column with missing values.
- Interactions: pairwise products of the 20 encoded columns most correlated with the label, skipping pairs from the same question and products that are almost always zero.

This gives about 1 230 columns without interactions and 1 410 with them.

III. Models

A. Baselines

The six required methods are implemented in implementations.py. As baselines, we use ridge regression (labels in $\{-1, 1\}$, $\lambda = 10^{-4}$) and regularized logistic regression with gradient descent ($\lambda = 10^{-4}$, $\gamma = 0.15$, 300 iterations) on the raw columns, mean-imputed and standardized.

B. Regularized logistic regression with Newton's method

We minimize the mean logistic loss with an L_2 penalty on all weights except the bias w_0:

$$L(w) = \frac{1}{n} \sum_{i=1}^{n} \left[\log\left(1 + e^{x_i^\top w}\right) - y_i \, x_i^\top w \right] + \lambda \sum_{j \geq 1} w_j^2. \tag{1}$$

Gradient descent needs a tuned step size and many iterations with 1 400 columns of very different scales. Newton's method uses the Hessian $H = \frac{1}{n} X^\top S X + 2\lambda I$, with $S = \mathrm{diag}(\sigma(x_i^\top w)(1 - \sigma(x_i^\top w)))$, and updates $w \leftarrow w - H^{-1} \nabla L(w)$. It converges in about 10 iterations. $X^\top S X$ is accumulated over blocks of rows so that memory stays bounded, and a step that increases the loss is halved. We use $\lambda = 3 \cdot 10^{-4}$.

C. Decision threshold

With 8.8% positives, few predicted probabilities exceed 0.5. We therefore choose the threshold t that maximizes F1 on the out-of-fold probabilities, by an exact search over all distinct scores (Fig. 1). The selected threshold is $t = 0.20$.

D. Validation protocol

All models are evaluated with the same stratified 5-fold cross-validation (seed 2026). For each fold, the feature encoder, the selection of interaction columns and the weights are fitted on the four training folds only. One threshold is chosen on the pooled out-of-fold probabilities, and we report the mean and standard deviation of the F1 of the five folds at that threshold.

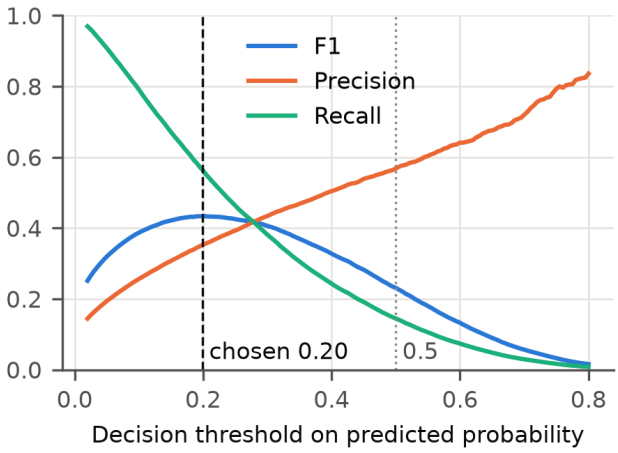


Figure 1. Out-of-fold F1, precision and recall of our model as a function of the decision threshold. F1 is 0.434 at the selected threshold and 0.231 at 0.5.

Table I
F1 of the positive class (5-fold CV, mean ± std). Raw: raw columns, mean-imputed and standardized.

Model	CV F1	Test F1
Ridge regression, raw	0.410 ± 0.004	
Logistic GD, raw	0.411 ± 0.003	
Newton, encoded features	0.432 ± 0.003	
Newton, encoded + interactions	0.434 ± 0.004	0.442

Our model is refitted on all labeled data with the same threshold.

IV. Results

Table I compares the baselines with each step of our model. Encoding the features adds +0.022, by far the largest gain. Interactions add +0.002, which is within one standard deviation across folds. The cross-validated F1 of the submitted model (0.434) is close to its test F1 on AIcrowd (0.442, accuracy 0.873), so the validation did not overfit.

A. What did not work

- Default threshold. At $t = 0.5$ our model predicts too few positives: F1 drops from 0.434 to 0.231 (Fig. 1).
- Changing the loss on raw features. Ridge regression and logistic regression on the same raw columns give the same F1 (0.410 vs 0.411): the bottleneck is the representation, not the model.
- More interactions. 12 interaction columns give 0.433 and 20 give 0.434. With 30 columns (about 430 products) the design matrix reaches 2.2 GB and did not fit in memory on our 8 GB machine. Plain Newton steps also diverged with a singular Hessian, because some products are almost always zero (incompatible answers). Removing these products and halving steps that increase the loss fixed the divergence, but we did not go beyond 20 columns.
- Tuning λ. Values from 10^{-5} to 10^{-3} all give F1 between 0.433 and 0.434, within one standard deviation: with 328k rows and 1 410 columns, regularization barely matters.

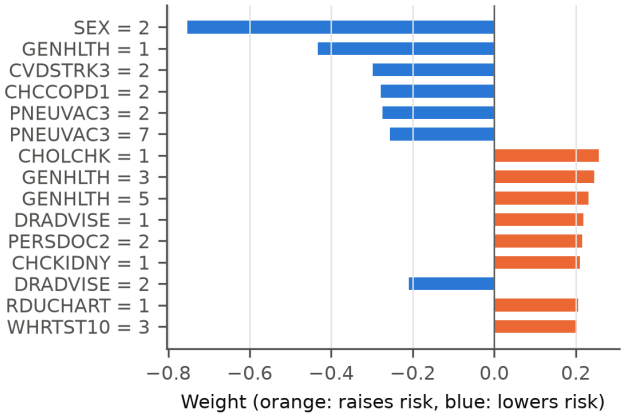


Figure 2. The 15 largest weights of our model.

V. Final Approach

VI. Discussion

The largest weights of our model (Fig. 2) match known risk factors. Female sex, excellent general health and the absence of stroke or COPD lower the predicted risk. Poor general health, kidney disease and having been advised by a doctor raise it. Since the weights of one-hot levels are relative to the reference level of the same question, they should be read as contrasts, not as absolute effects.

VII. Conclusion